	$P = Fv = (1642.9)(36.84) = 60.5$ kW
8	**Answer: A** When both chambers are in pressure and thermal equilibrium, $\frac{P_1}{T_1} = \frac{n_1 R}{V_1} = \frac{P_2}{T_2} = \frac{n_2 R}{V_2}$ $$\frac{n_1 R}{V_1} = \frac{n_2 R}{V_2};\ \frac{n_1}{n_2} = \frac{V_1}{V_2} = \frac{AL_1}{AL_2} = \frac{L_1}{L_2}$$ where A is the cross-sectional area of the cylinder. Hence the ratio of the lengths is equal to the ratio of the number of mol of gases $$\frac{n_1}{n_2} = \frac{\frac{20 \times 10^{-3}}{28}}{\frac{45 \times 10^{-3}}{4}} = 0.063$$
9	**Answer: C**